Le réseau de transport d'électricité

Assessment of the potential of Earth observation data and deep convolutional neural networks to improve the estimation and forecast of the solar power production in France

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Not all PVs are measured equally

PV Energy

Plants ($\geq 1\ MWp$)

Mesurable : RTE has access to real time measurements of the output

Decentralized $< 1\ MWp$)

Small plants

Distributed PV

- Increasing installed capacity
- Growing impact on the grid
- Unknown output

Increased uncertainty managing the balancing between supply and demand

Bridging the gap leveraging a regional PV model and a registry

We can fit a **regional PV Model** to get an accurate estimation of the decentralized PV
However, **we need statistics on the characteristics of PV systems**

Regional PV model

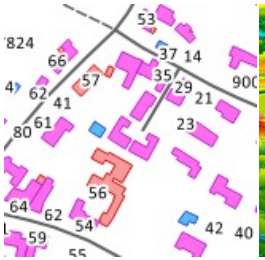
Distributed PV chatacteristics cannot be extrapolated from plants

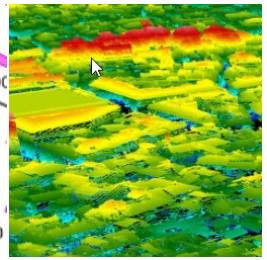
Construct a **registry** containing the following characteristics :

- **Localization**
- **Tilt**
- **Azimuth**
- **Surface area**
- **Installed capacity**

Leveraging deep learning and EO data to build the registry

Previous studies (Yu et al, 2018, Mayer et al, 2022) proved that **deep learning** and **Earth observation data** can be leveraged to **construct large scale PV registries**





Id	Loc.	Tilt	Azim	Inst. Cap.	Area
0	45°, 3°	30°	-10°	3 kWp	12 sqm
1	44°, 3°	45°	-30°	2,9 kWp	10 sqm
2	46°, 3°	10°	0°	30 kWp	100 sqm
			…		

1. Detection and segmentation of PV installations

➔ **Image classification and image segmentation**

2. Filtering and characteristics extraction using topological and LiDAR data

➔ **Automatic extraction of the characteristics**

Y. Tremenbert's internship

Outcome : automated PV registry

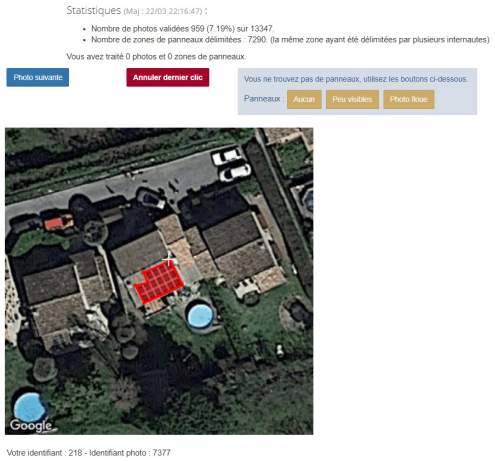
➔ **Aggregated and standardized data**

➔ Expected to be the **largest registry** with this level of detail

Deep learning needs data: the BDAPPV crowdsourcing campain

Existing models cannot be transferred to new locations

BDAPPV
BASE DE DONNÉES D'APPRENTISSAGE PROFOND PV
Un outil aux service de tous.



- We need **training data** for the classification and segmentation models

The crowdsourcing campain allowed to **identify nearly 10k installations**.

Currently, we are **delineating the installations**

Feel free to contribute !

- We need **calibration data** to calibrate the registry

Leverage the **data on small power plants** provided by **BDPV** and manually annotated tiles to get **ground truth data on the localization and characteristics** of the installations.

https://www.bdpv.fr/_BDapPV/index.php

BDPV